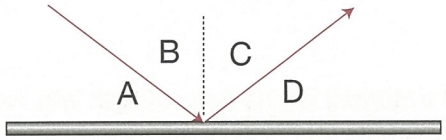
5.6.4 Light Energy

After completing this chapter you should be able to apply models, theories and laws to situations involving energy, force and motion.

For Questions 1 to 10 choose the best answer (A), (B), (C) or (D).

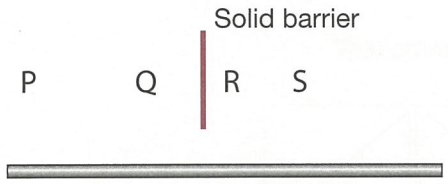
1. Consider the following diagram showing a ray of light reflecting from a surface.



Which angle is the angle of incidence?

(A) A
(B) B
(C) C
(D) D

2. Consider four people P, Q, R and S sitting in front of a mirror as shown:



Which statement is *false*?

(A) P can see an image of S in the mirror.
(B) S can see an image of Q in the mirror.
(C) Q can see an image of R in the mirror.
(D) R can see an image of S in the mirror.

3. Which of these is *not* caused by refraction of light?

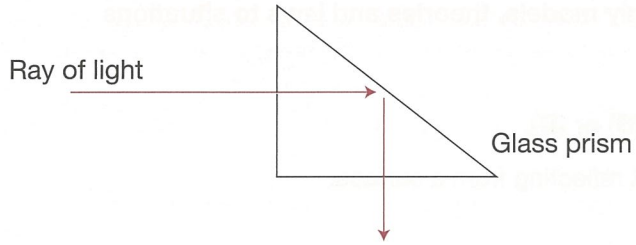
(A) A lens can focus light.
(B) A magnifying glass makes things look bigger.
(C) A pencil looks bent in a glass of water.
(D) You can see yourself in a mirror.

4. The path of a ray of light bends when it enters water at any angle other than 90°. What is this property of light called?

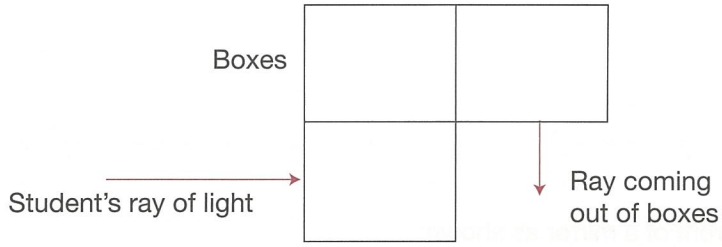
(A) Absorption.
(B) Diffraction.
(C) Reflection.
(D) Refraction.

Use the following information to answer the next TWO questions.

The diagram shows a ray of light passing into and out of a prism of glass.

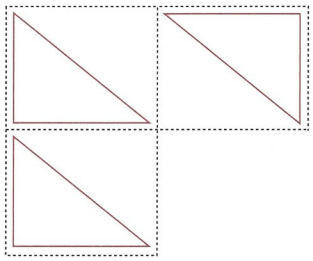


A teacher set up some 'black boxes' that contained glass prisms and a student shone a ray of light into the first box. The path of the ray she observed coming out of the boxes is shown below.

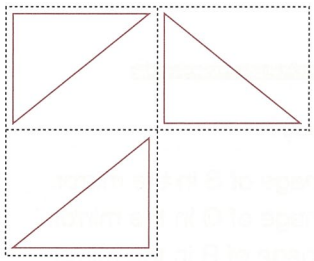


5. Which choice correctly shows the orientation of the prisms in the boxes?

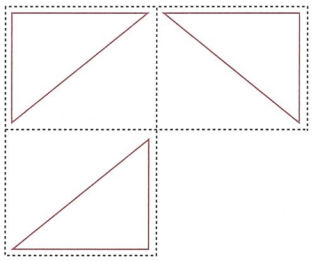
(A)



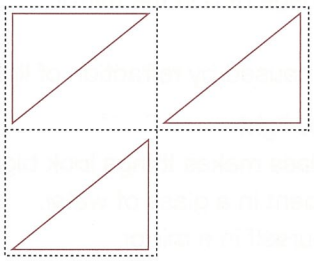
(B)



(C)



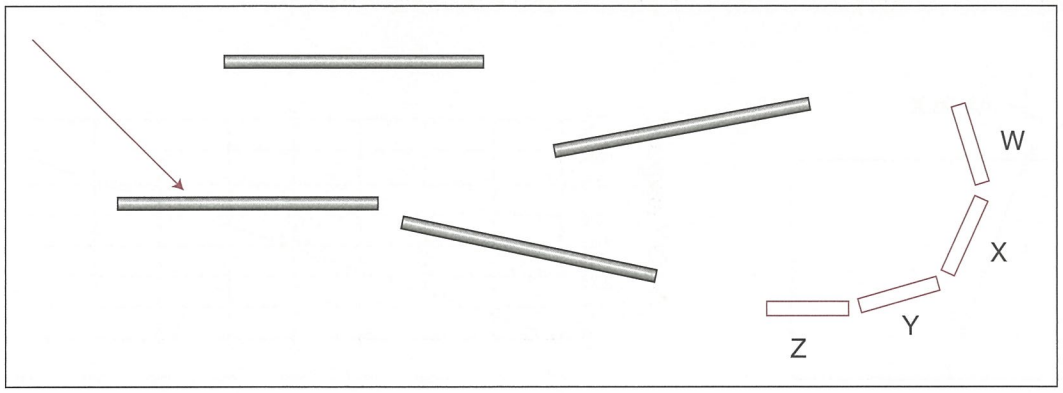
(D)



6. Which principle of physics is being shown here?

(A) During reflection, the angle of incidence is equal to the angle of reflection.
(B) During reflection, the angle of incidence is equal to the angle of refraction.
(C) During refraction, the angle of incidence is equal to the angle of reflection.
(D) During refraction, the angle of incidence is equal to the angle of refraction.

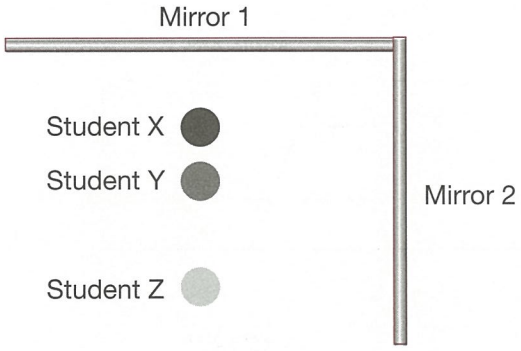
7. Consider the diagram which shows a ray of light heading for the first of several mirrors.



Which target will the ray of light hit?

(A) W
(B) X
(C) Y
(D) Z

8. The diagram shows three students standing near two mirrors placed at right angles to each other.

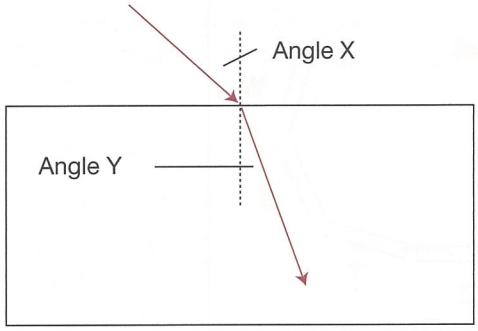


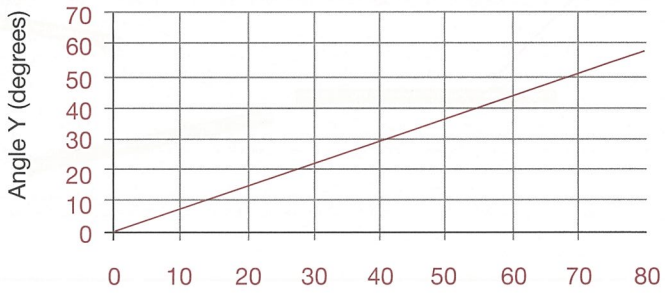
Which statement is correct?

(A) Student X can see both students Y and Z in both mirrors 1 and 2.
(B) Student X can see students Y and Z in mirror 2, but neither student in mirror 1.
(C) Student Y can see both students X and Z in both mirrors 1 and 2.
(D) Student Z can see students Y and Z in mirror 1 but neither student in mirror 2.

Use the following information to answer the next TWO questions.

The diagram shows what happens when a ray of light goes into a solid plastic block.



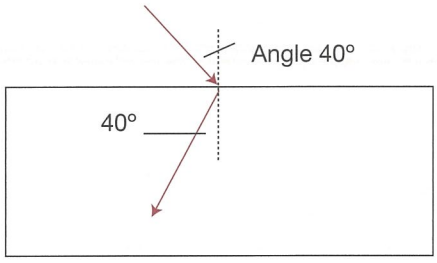


9. When angle X is 35°, what will be the value of angle Y?

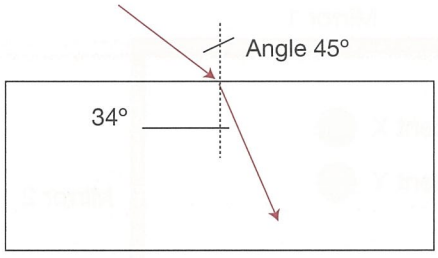
(A) 32°
(B) 34°
(C) 36°
(D) 38°

10. Which diagram shows the correct path for the beam of light?

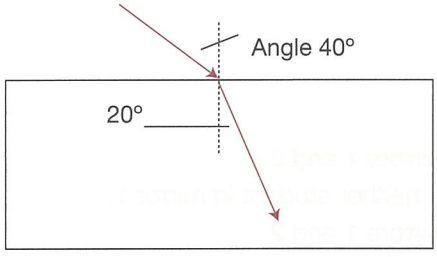
(A)



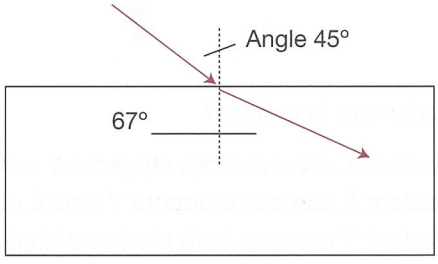
(B)



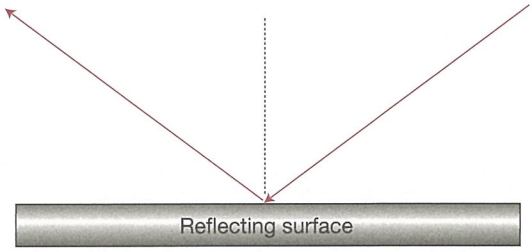
(C)



(D)

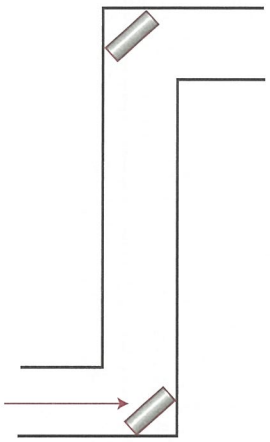


11. On the diagram, label the incident ray, the reflected ray, the angle of incidence, the angle of reflection and the normal to the surface.

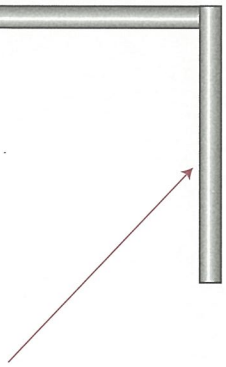


12. Draw in the rest of the path of the rays of light in the diagrams below. All surfaces are reflecting surfaces.

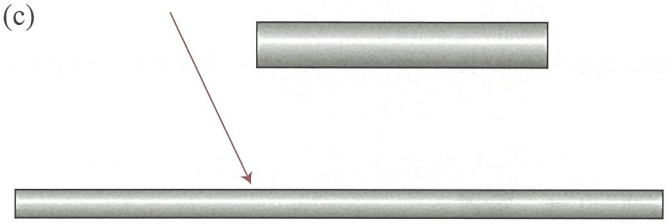
(a)



(b)



(c)



13.

(a) Draw a diagram to show how the word 'EMERGENCY' should appear as a sign on top of a rescue vehicle such as a fire engine, ambulance or police rescue vehicle.

(b) Explain why it should appear this way.

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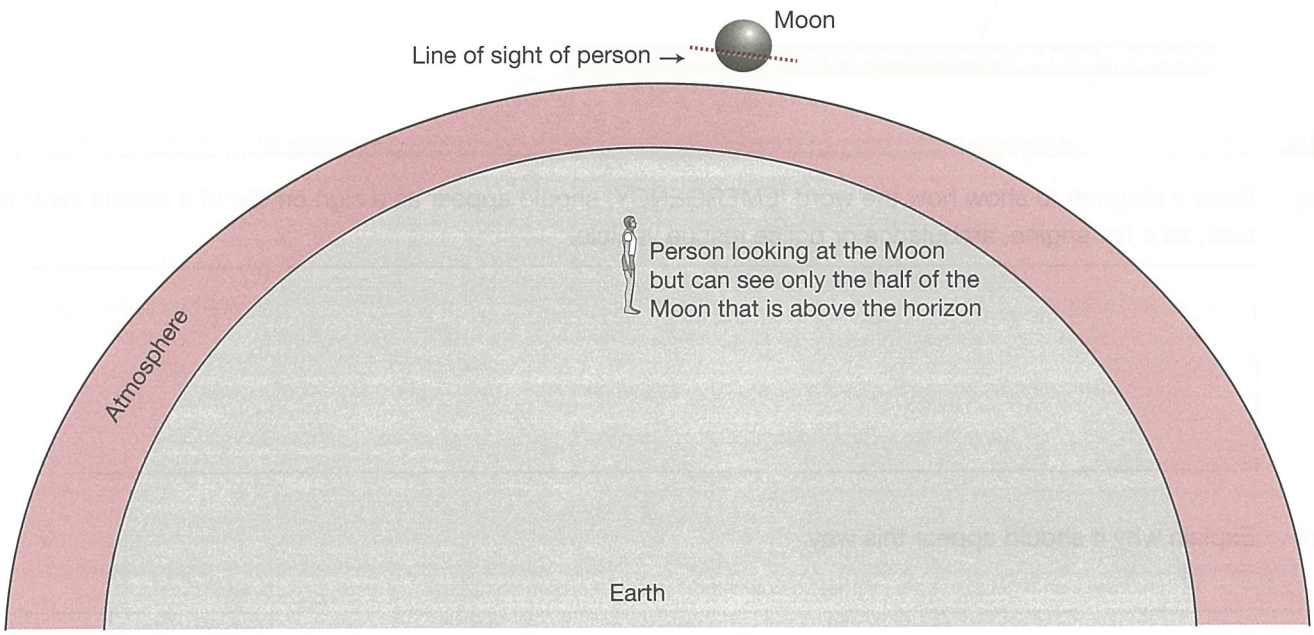
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14. With the aid of a labelled diagram, show what is meant by refraction of light.

15.

(a) Complete the diagram to show how the Moon appears larger than normal when it is near but not yet fully above the horizon.



(b) Explain, using your own words, why the Moon on the horizon looks bigger than usual.

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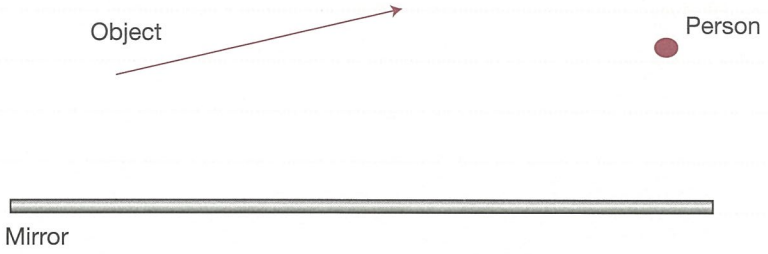
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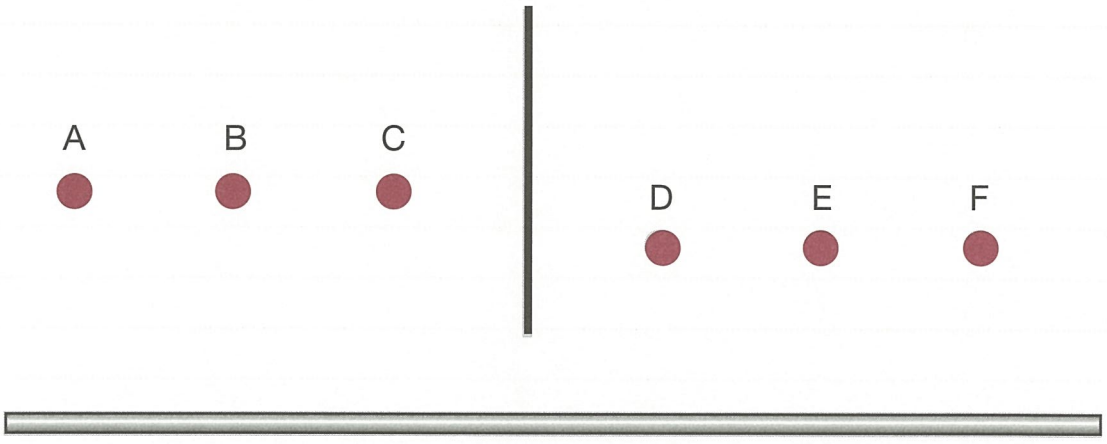
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16. The diagram shows an object in front of a mirror. Draw an accurate diagram to determine how a person looking from point ● would see the image of the object in the mirror.



17. The diagram shows several people (●) sitting in front of a mirror. Who can see whom?



A can see …	D can see …
B can see …	E can see …
C can see …	F can see …

Notes